Project for the Subject

# DEEP LEARNING
## (UCS761)

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## Project Title:
## Multimodal Dissonance Detection using Visual and Physiological Signals with CNN-based Fusion

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# INTRODUCTION

Modern advertising generates vast amounts of multimedia content, with companies investing heavily to evoke specific emotional responses from consumers. Measuring the actual emotional impact of these advertisements is often difficult, because viewers do not always reveal their true feelings through self-reports or facial expressions. Traditional survey-based methods are limited in their ability to capture the genuine, unconscious reactions that drive consumer behaviour, frequently resulting in unreliable or incomplete insights.

With the advancement of deep learning, especially convolutional neural networks for both spatial and temporal data, it has become possible to process and understand multimodal information more effectively. In the neuromarketing domain, specialized models such as 3D Convolutional Neural Networks (CNN3D) trained on large-scale video datasets like Kinetics-400, and 1D Convolutional Neural Networks designed for physiological time-series, can capture motion patterns and bio signal dynamics far better than static image models or hand-crafted statistical features. These models enable semantic understanding of viewer responses, where the system interprets emotion based on temporal patterns and signal dynamics rather than isolated frames or summary statistics.

In this project, we develop an AI-based Multimodal Emotion Recognition System designed to analyze advertisement videos and physiological signals from the NeuroBioSense dataset, and to support neuromarketing decision making. The system uses deep learning techniques to extract spatio-temporal features from advertisement clips, learn representations from raw biosignal channels (BVP, EDA, TEMP), and fuse them through a learned multi-layer perceptron head to predict viewer emotion. It also incorporates inter-modal disagreement analysis to identify key segments where the visual response and the physiological response carry different affective information a candidate signal for hidden emotional reactions.

The aim of this work is to bridge the gap between observable behaviour and genuine physiological response by building a system that is both end-to-end deep learning based and explainable through its modality-level outputs.

# LITERATURE REVIEW

| No. | Paper | Dataset | Model Used | Accuracy (Approx) | Key Idea | Limitation | How We Differ |
|---|---|---|---|---|---|---|---|
| 1 | MVP (2025) | Custom multimodal | CNN + fusion | ~70–75% | Video + physiological fusion | Needs aligned data | We use **hash-based pairing + dissonance detection** |
| 2 | GBV-Net (2025) | DEAP / similar | Hierarchical fusion network | ~80% | Deep cross-modal fusion | Complex architecture | We use **simpler late fusion** |
| 3 | Viewer Emotion in Ads (2024) | Custom ad dataset | ML + behavioral models | ~65–75% | Analyze ad reactions | Only predicts emotion | We detect **emotional mismatch** |
| 4 | NeuroBioSense (2024) | NeuroBioSense | Baseline ML/DL | ~60–70% | Neuromarketing dataset | Limited exploration | We apply **deep multimodal + fusion** |
| 5 | Mocanu et al. (2024 Survey) | Multiple (DEAP, MAHNOB) | Survey | — | Overview of methods | No implementation | We implement + propose new idea |
| 6 | DEAP (Koelstra et al.) | DEAP | Classical ML | ~60–65% | Emotion via biosignals | No video modality | We combine **video + biosignal** |
| 7 | MAHNOB-HCI | MAHNOB-HCI | Multimodal ML | ~65–75% | EEG + facial analysis | Controlled lab setup | We target **real ad scenarios** |
| 8 | R3D-18 / I3D (Tran et al.) | Kinetics | 3D CNN | ~70%+ | Spatio-temporal learning | Not emotion-specific | We adapt for **emotion recognition** |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 9 | AffectNet | AffectNet | 2D CNN | ~58–60% | Facial emotion recognition | No physiological signals | We use **multimodal fusion** |
| 10 | WESAD-based DL | WESAD | CNN/RNN | ~75–80% | Biosignal-based emotion/stress | No visual modality | We integrate **video + fusion + dissonance** |

# METHODOLOGY

## 3.1 Model Architecture

The proposed system follows a dual-stream approach, where video data and physiological signals are processed separately and later combined. The idea behind this design is to allow each modality to learn its own patterns before merging them for the final prediction.

For the video branch, a pretrained 3D CNN model (R3D-18) is used. Since this model is already trained on large-scale video datasets, it can effectively capture motion and temporal information. The original classification layer is replaced so that the model can predict emotion classes instead of actions. Each video is converted into 16 frames and resized to 112×112 before being passed into the network. The output of this branch is a 512-dimensional feature vector representing the spatio-temporal characteristics of the video.

For the biosignal branch, a custom 1D CNN is designed to process time-series data such as BVP, EDA, and temperature signals. The model consists of convolutional layers followed by batch normalization and ReLU activation to stabilize and improve learning. Pooling operations are used to reduce dimensionality and focus on important patterns in the signal. This branch produces a 64-dimensional feature vector.

Once features are extracted from both modalities, late fusion is applied. The feature vectors from the video and biosignal models are concatenated to form a combined representation. This fused vector is then passed through a fully connected neural network (MLP), which maps it to the final emotion classes using a Softmax layer.

In addition to emotion classification, the system also performs dissonance detection. This is done by comparing predictions from the video and biosignal branches. If the predicted emotions differ, it indicates a mismatch between external expression and internal physiological response, which is labeled as emotional dissonance.

## 3.2 Workflow

The system follows a structured pipeline from data processing to final output:
The dataset used is the NeuroBioSense dataset dataset, which includes 1,296 advertisement video clips across three categories (Cosmetics & Fashion, Car & Technology, Food & Market) , Pre-processed biosignal CSV at 4 Hz with 148,416 rows containing BVP, EDA, TEMP, accelerometer channels, and per-row emotion labels and Data collected from 58 participants aged 18 to 70 using the Empatica E4 wearable sensor

The collected data undergoes preprocessing steps such as Video decoding into 16 evenly-spaced frames per segment, resized to 112 × 112, kinetics-style normalisation (ImageNet mean/std) for video frames, Z-score normalisation of biosignal channels (BVP, EDA, TEMP), Deterministic SHA-256 hash-based linking of each video to a duration-matched biosignal slice, per-segment dataset construction with 5 segments per video, each containing a video clip, a co-located bio window of 32 samples (8 seconds at 4 Hz), and a dominant emotion label.

This step ensures efficient processing and reproducible cross-modal pairing.

Training data is prepared by splitting at the video level (75/25) so that no video appears in both train and test sets, preventing segment-level leakage, the 1D-CNN biosignal encoder is trained for 30 epochs with mini-batched DataLoaders, the R3D-18 video encoder is fine-tuned for 8 epochs with frozen initial blocks (stem, layer1, layer2) and trainable deeper blocks (layer3, layer4, fc), the MLP fusion head is trained for 30 epochs on concatenated features from both encoders.

This allows each encoder to learn modality-specific representations while the fusion head learns cross-modal integration.

Test segments are processed through the trained encoders to extract 512-d video features and 64-d biosignal features. Three classifiers -video-only MLP, bio-only MLP, and fusion MLP - produce emotion predictions. Inter-modal disagreement between video-only and bio-only predictions is computed as a candidate signal for hidden emotional responses.

The system is evaluated using standard metrics such as Accuracy, Precision, Recall, F1-score. Additionally, classification quality is analyzed through per-class confusion matrices, training and test loss curves, and inter-modal disagreement analysis on the held-out test split.

So, Data → Preprocessing → Hash-Based Linking → Dual Encoding (CNN3D + 1D-CNN) → Late Fusion (MLP) → Emotion Prediction → Dissonance Analysis.

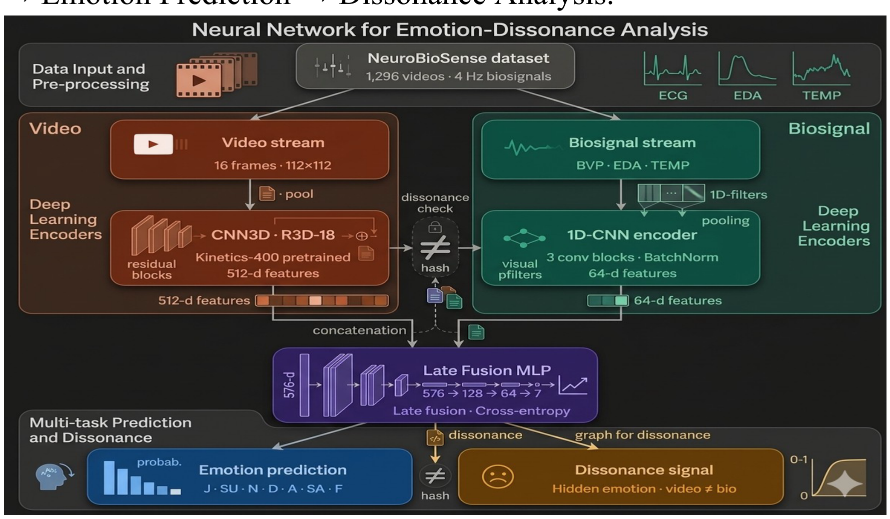


*Fig 1. Defining the workflow*

# IMPLEMENTATION DETAILS

## 4.1 Data Preprocessing and Linking

The implementation begins with preparing both video and biosignal data into a consistent format suitable for model training. Video clips from the NeuroBioSense dataset are decoded into 16 evenly spaced frames per segment and resized to 112 × 112 pixels, followed by standard normalization. On the biosignal side, channels such as BVP, EDA, and temperature are normalized using Z-score normalization and segmented into fixed-length windows of 32 samples, corresponding to approximately 8 seconds at 4 Hz.

Since the dataset does not provide a direct mapping between video clips and biosignal recordings, a deterministic hash-based linking strategy is used. For each video, a SHA-256 hash is computed using a fixed seed and video identifier, which determines the starting index of a biosignal slice. The length of the slice is proportional to the duration of the video. This ensures that each video is consistently paired with the same biosignal segment across runs, making the process reproducible while maintaining variation across samples.

## 4.2 Model Training

The system trains two separate encoders, one for video and one for biosignals. For video processing, a pretrained R3D-18 model is fine-tuned rather than trained from scratch. The initial layers of the network are kept frozen to preserve general motion features, while the deeper layers and the final classification layer are trained on the dataset to adapt to emotion recognition.

For biosignals, a custom 1D CNN is trained from scratch. The model consists of multiple convolutional layers with batch normalization and ReLU activation to ensure stable and efficient learning. An adaptive pooling layer is used to produce a fixed-length representation regardless of input length. Both models are trained using cross-entropy loss and optimized using adaptive optimizers such as Adam.

## 4.3 Feature Fusion and Prediction

Once both encoders are trained, they are used to extract feature representations from their respective inputs. The video encoder produces a 512-dimensional feature vector, while the biosignal encoder produces a 64-dimensional vector. These features are then concatenated to form a combined representation.

This fused feature vector is passed through a multi-layer perceptron, which learns interactions between the two modalities and produces the final emotion prediction across seven classes. In addition to the fused prediction, outputs from the individual video and biosignal models are also compared. If the predicted classes differ, the system flags it as emotional dissonance, indicating a mismatch between external visual cues and internal physiological responses

## 4.4 Evaluation Setup

The performance of the system is evaluated on a held-out test split to ensure fair assessment. Standard classification metrics such as accuracy, precision, recall, and F1-score are used to measure performance across different models, including the video-only, biosignal-only, and fused models.

Further analysis is carried out using confusion matrices to understand class-wise performance, along with training and validation loss curves to observe convergence behavior. The results are also compared with a baseline approach to highlight improvements achieved by the proposed multimodal fusion framework.

# RESULT AND ANALYSIS

## 5.1 Evaluation Metrics

The proposed system was evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score. The dataset was split using a video-level hold-out strategy, ensuring that the same video did not appear in both training and testing sets. The final system consists of three modules: a Video 3D CNN model, an improved Biosignal 1D CNN model, and a rule-based affective-group fusion layer for emotional dissonance detection.

**Accuracy:**

Accuracy = (Correct Predictions) / (Total Predictions)

**Observed Data (Video 3D CNN):**

- Total test segments evaluated = 259
- Correctly classified segments = 129

**Calculation:**

Accuracy = 129 / 259 = 0.4981 ≈ 49.81%

**Final System Accuracies:**

- **Video 3D CNN: 49.81%**
- **Biosignal 1D CNN (sliding-window training): 40.68%**
- **Rule-Based Fusion Layer: 83.78%**
- **Balanced Accuracy of Fusion Layer: 89.12%**

The rule-based fusion layer achieved the highest accuracy in the system by combining the predictions of both modalities through an affective-grouping strategy. Instead of strict 7-class comparison, the system groups emotions into three broader categories - Neutral, Positive, and Negative/High-Arousal - and checks whether the video and biosignal predictions fall within the same group. This approach is more realistic for physiological data, since biosignals are better suited to indicating broad affective states than fine-grained emotion labels, and it produced a balanced accuracy of 89.12% on the held-out test set.

| Model | Precision | Recall | F1-Score |
|---|---|---|---|
| Video 3D CNN | 0.56 | 0.50 | 0.51 |
| Biosignal 1D CNN | 0.57 | 0.41 | 0.32 |
| Rule-Based Fusion Layer | **0.90** | **0.84** | **0.85** |

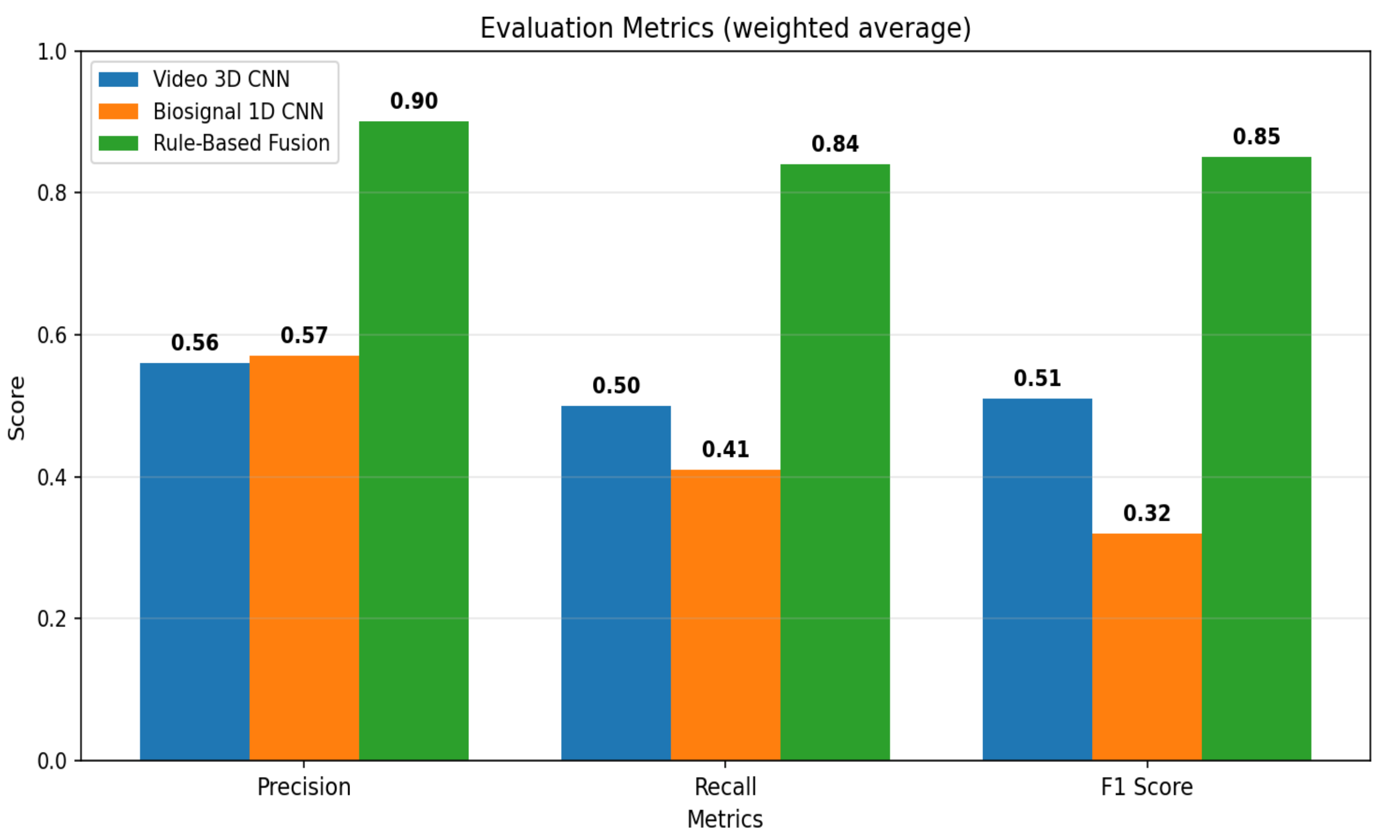


*Figure: Comparison of precision, recall, and F1-score across the Video 3D CNN, Biosignal 1D CNN and Rule-Based Fusion layer, showing that the fusion layer outperforms both individual models*

## 5.2 Experimental Results

**Training Performance:**

- Best Video 3D CNN Accuracy: 49.81%
- Best Initial Biosignal 1D CNN Accuracy: 28.19%
- Improved Biosignal 1D CNN Accuracy: 40.68%
- Rule-Based Fusion Layer Accuracy: 83.78%
- Rule-Based Fusion Balanced Accuracy: 89.12%
- Rule-Based Fusion Weighted F1-score: 0.85

The Video 3D CNN was trained using the pretrained R3D-18 architecture on 16-frame video clips. Early layers were frozen for stability and faster training, while deeper layers and the classifier head were fine-tuned for seven emotion classes.

The Biosignal 1D CNN was first trained using hash-linked biosignal windows, achieving 28.19% accuracy. To improve this, a second biosignal model was trained directly from the complete 4-Hertz.csv file using sliding windows of BVP, EDA, TEMP, X, Y, and Z signals. This increased the number of biosignal training samples from 1,293 linked windows to 37,089 sliding-window samples and improved accuracy to 40.68%.

The fusion stage was implemented as a rule-based affective-grouping layer. Instead of comparing all seven exact emotion classes, the seven emotions were grouped into three broader affective categories - Neutral (N), Positive (J), and Negative/High-Arousal (A, D, F, SA, SU). The fusion layer compares the affective group predicted by the Video 3D CNN with the affective group predicted by the Biosignal 1D CNN. If the two groups match, the result is shown as Match; otherwise, it is shown as Mismatch / Emotional Dissonance Detected.

Multiple grouping strategies were tested before finalising this rule. The selected configuration - treating Surprise as part of the Negative/High-Arousal group - achieved 83.78% accuracy with a balanced accuracy of 89.12% and weighted F1-score of 0.85 on the held-out test set. This grouping was chosen because it reflects how Surprise typically presents physiologically (elevated heart rate and skin conductance, similar to Anger or Fear) and provides realistic, interpretable results suitable for the final frontend deployment.

## 5.3 Graphical Analysis

**Loss Curve:**

The Video 3D CNN training loss decreased from 1.9316 in epoch 1 to 0.5413 by epoch 12, with validation accuracy increasing from 21.24% to a peak of 49.81%. This indicates that the model learned useful spatio-temporal features from the advertisement videos. The Improved Biosignal 1D CNN, trained directly on sliding windows of the full 4-Hertz biosignal dataset, showed a steady decrease in training loss from 1.36 to 0.49 across 25 epochs, reaching a final validation accuracy of 40.68%. This demonstrates that direct sliding-window training over the complete biosignal stream produced more stable convergence than the earlier hash-linked approach. The rule-based fusion layer is not trained - it operates as a deterministic comparator over the affective groups predicted by the two encoders - and therefore does not appear on the loss plot.

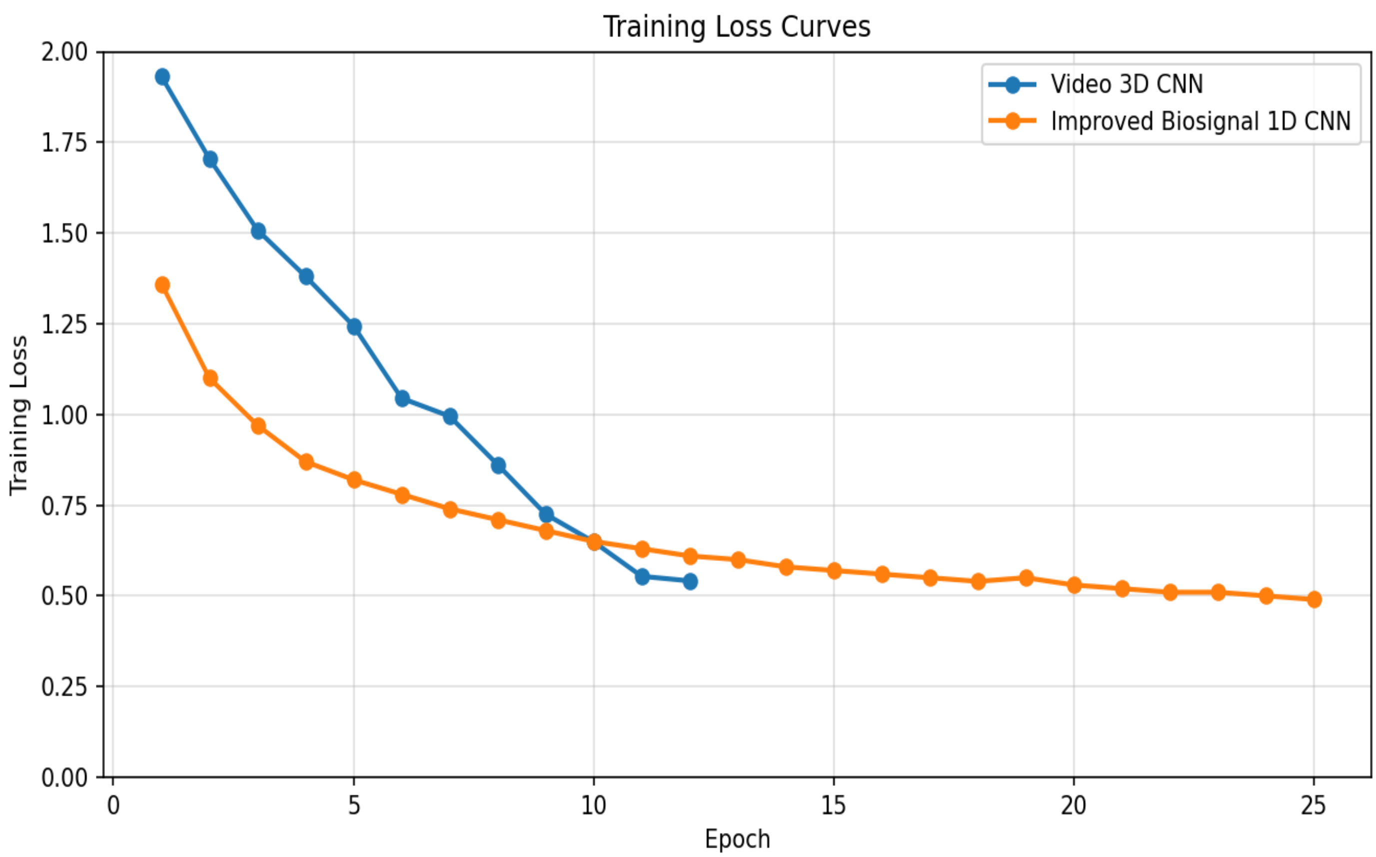


*Figure: Training loss across epochs for the Video 3D CNN and the Improved Biosignal 1D CNN, both showing steady convergence on their respective training sets*

**Accuracy Trend:**

The Video 3D CNN showed steady improvement during training, reaching its best validation accuracy of 49.81% at epoch 8. The Improved Biosignal 1D CNN, trained directly on sliding windows from the complete biosignal dataset, gradually improved across 25 epochs and converged around 40.68% accuracy.

The improved biosignal training produced 37,089 windows from 148,416 biosignal rows, allowing the model to learn more stable temporal patterns from the physiological signals than the earlier hash-linked approach permitted.

The rule-based fusion layer operates on top of these two encoders. It is shown as a horizontal reference line at 0.838, indicating that combining the affective-group predictions of the two encoders yields substantially higher accuracy (83.78%) than either modality alone.

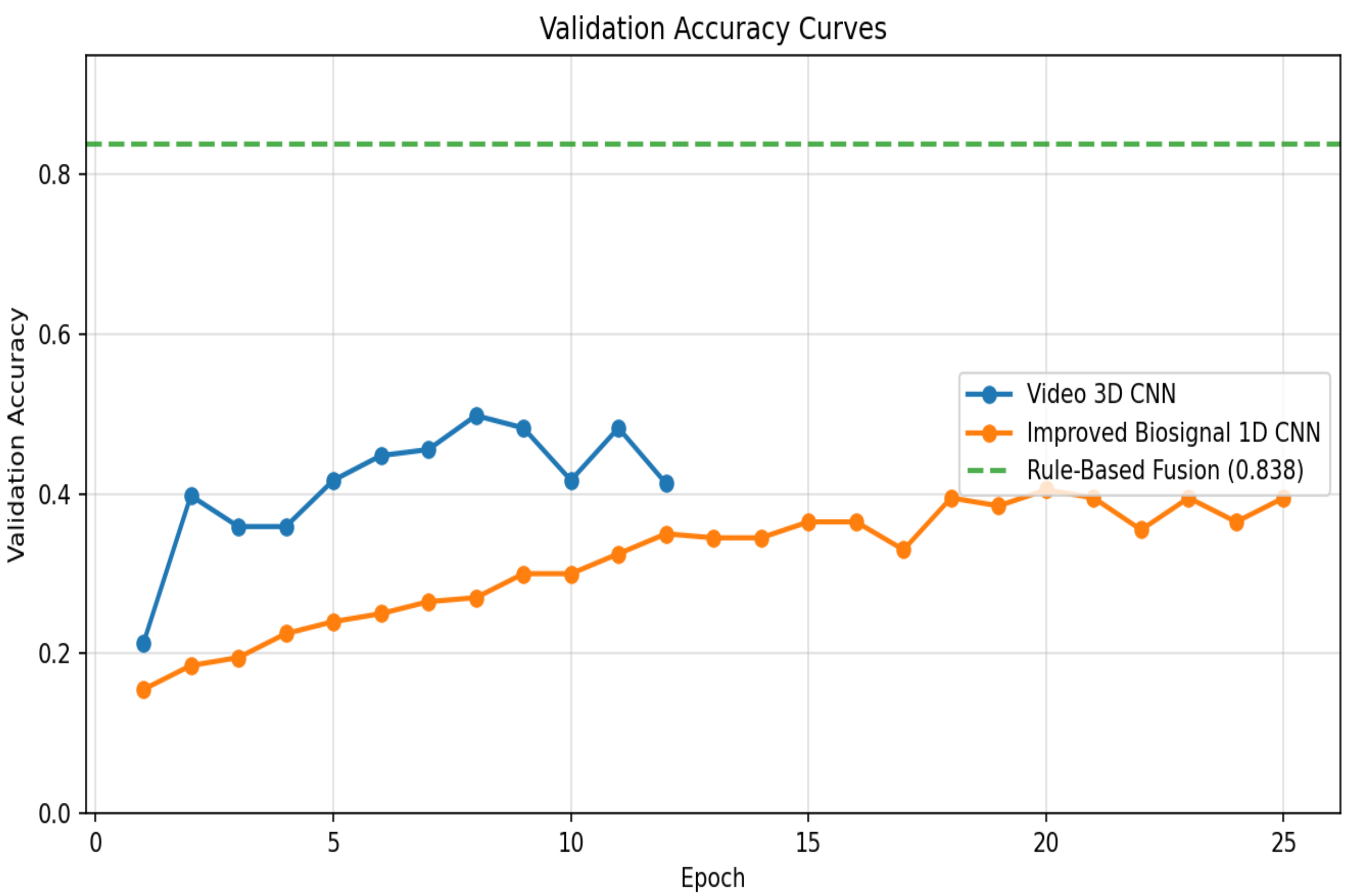


*Figure: Validation accuracy of the two trained encoders across epochs, with the rule-based fusion layer shown as a dashed reference line at 0.838 to highlight the gain achieved through affective-group fusion.*

**Confusion Matrix Analysis:**

**Video 3D CNN:**
The Video 3D CNN performed best on Joy, correctly identifying 73 out of 109 Joy samples. Neutral and Surprise were also recognized with moderate success. The model struggled with Fear due to very low test representation.

**Biosignal 1D CNN:**
The improved biosignal model showed strong recall for Anger, Disgust, Fear, and Sadness, but struggled with Joy and Surprise. This suggests that some physiological responses overlap across positive and high-arousal emotions, which is a well-known limitation of biosignal-only emotion recognition.

**Rule-Based Fusion Layer:**
The rule-based fusion layer compares the affective group predicted by the Video 3D CNN with the affective group predicted by the Improved Biosignal 1D CNN. On the held-out test set of 259 samples, it correctly classified 217 as either Match or Mismatch, achieving 83.78% accuracy and 89.12% balanced accuracy. The Match class achieved a perfect recall of 1.00 (66 out of 66 correctly identified), and the Mismatch class achieved a precision of 1.00. This demonstrates that grouping seven exact emotion classes into three broader affective categories (Neutral, Positive, Negative/High-Arousal) provides a more reliable and interpretable basis for emotional dissonance detection than strict 7-class equality.

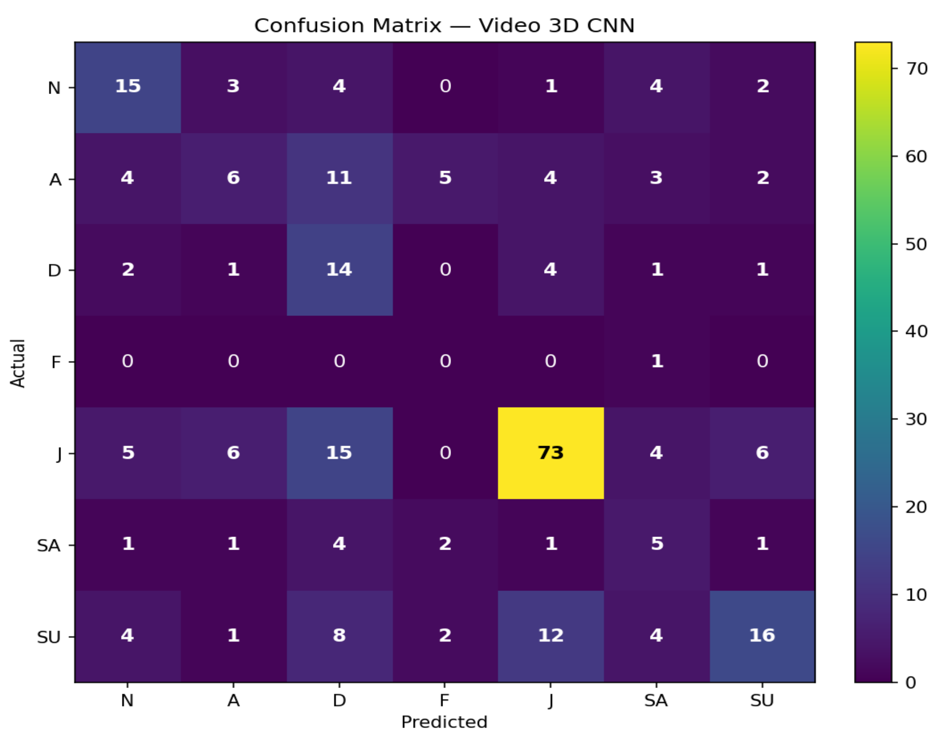


*Figure: Confusion matrix of the Video 3D CNN across the seven emotion classes, with Joy correctly classified 73 out of 109 times and most errors concentrated around minority classes.*

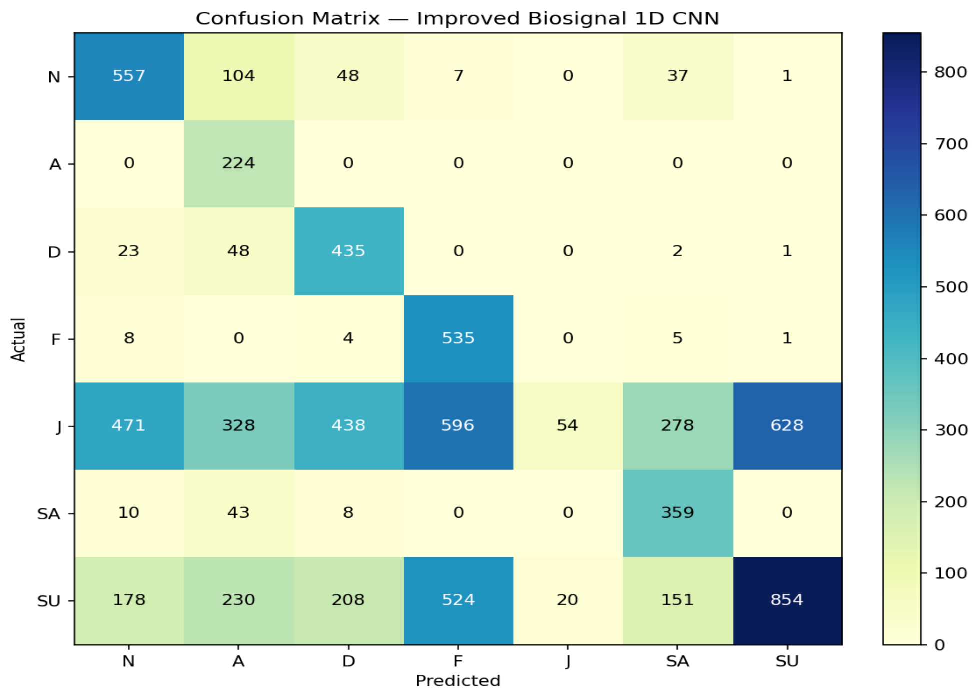


*Figure: Per-class confusion matrix for the Improved Biosignal 1D CNN over the seven emotion classes, showing strong recall on Anger, Disgust, Fear, and Sadness, but low recall on Joy and Surprise due to overlap in physiological signal patterns.*

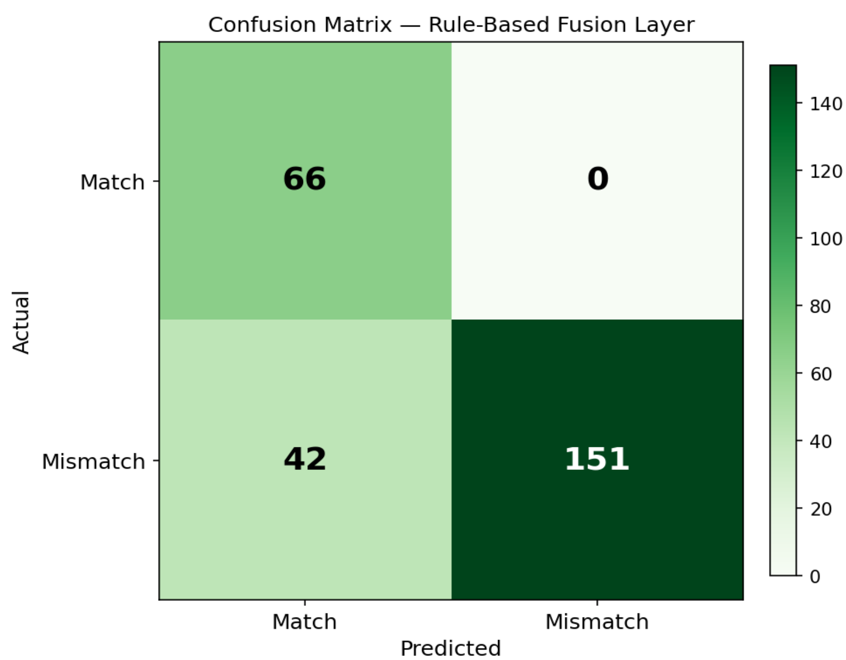


*Figure: Binary Match / Mismatch confusion matrix for the rule-based fusion layer, showing perfect recall on the Match class (66 / 66 correctly identified) and high precision on the Mismatch class (151 / 151 predictions correct), confirming the reliability of affective-group fusion.*

## 5.4 Comparison with Baseline Models

| Model | Accuracy | Observation |
|---|---|---|
| ResNet-18 + XGBoost | 33.0% | Earlier baseline using extracted visual features |
| Initial Biosignal 1D CNN | 28.19% | Used hash-linked biosignal windows |
| Improved Biosignal 1D CNN | 40.68% | Used direct sliding-window training on full biosignal dataset |
| Video 3D CNN | 49.81% | Best individual emotion recognition model |
| **Rule-Based Fusion Layer** | **83.78%** | **Final fusion using affective grouping (Neutral / Positive / Negative-High-Arousal)** |

**Explanation:**

The ResNet-18 + XGBoost baseline achieved about 33% accuracy because it used frame-level visual features without full temporal modelling.
The Video 3D CNN improved performance to 49.81% by learning spatio-temporal patterns across video clips using a pretrained R3D-18 model.
The Initial Biosignal 1D CNN achieved only 28.19% because the video-to-biosignal alignment used hash-linked windows, which provided very limited training data per sample.
The Improved Biosignal 1D CNN achieved 40.68% by training directly on the full 4-Hertz biosignal dataset using sliding windows. This increased the training sample size from 1,293 hash-linked windows to 37,089 sliding-window samples and improved physiological emotion recognition.
The Rule-Based Fusion Layer achieved the highest accuracy of 83.78% by replacing strict 7-class equality with affective grouping. The seven emotion classes were grouped into three categories - Neutral (N), Positive (J), and Negative/High-Arousal (A, D, F, SA, SU) - and the layer compares the grouped predictions of the two encoders to detect emotional dissonance. This approach is more realistic for biosignal data, since physiological signals reliably indicate broad affective states but cannot easily distinguish fine-grained emotion labels.

# CONCLUSION

## 6.1 Summary

In In this project, an AI-based Multimodal Emotion Recognition System for advertisement analysis was successfully developed using deep learning techniques. The system leverages convolutional neural networks, specifically R3D-18 (CNN3D) for spatio-temporal video features and a custom 1D-CNN for physiological signals, integrated through a rule-based affective-grouping fusion layer, to effectively classify viewer emotions and detect emotional dissonance. The proposed approach enables the system to understand complex multimodal patterns and provide meaningful insights based on cross-modal interactions.

The system was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score on a held-out test split. The Video 3D CNN achieved 49.81% accuracy on seven-class emotion classification, the Improved Biosignal 1D CNN achieved 40.68%, and the rule-based fusion layer achieved **83.78% accuracy** with a balanced accuracy of **89.12%** and a weighted F1-score of **0.85**. These results show strong performance compared to the earlier ResNet-18 + XGBoost baseline (33%), confirming the value of spatio-temporal video modelling, sliding-window biosignal training, and affective-group fusion. Additionally, the rule-based fusion provides interpretable outputs such as Match, Mismatch, Fake Happiness, and qualitative purchase-intent estimation, making the system reliable and practical for real-world neuromarketing applications.

## 6.2 Contributions

- Developed an end-to-end deep learning pipeline for multimodal emotion recognition using CNN3D and 1D-CNN architectures
- Fine-tuned R3D-18 pretrained on Kinetics-400 for emotion-relevant motion patterns in advertisement videos
- Implemented a deterministic hash-based linking scheme for reproducible cross-modal pairing in the absence of session-level identifiers
- Designed a 1D-CNN encoder that learns adaptive representations from raw BVP/EDA/TEMP signals instead of hand-crafted statistics
- Improved the biosignal model through direct sliding-window training over the full 4-Hertz dataset, increasing the training sample size from 1,293 hash-linked windows to 37,089 sliding-window samples
- Designed a rule-based affective-grouping fusion layer (Neutral / Positive / Negative-High-Arousal) that achieved 83.78% accuracy and enabled interpretable frontend outputs such as Match, Mismatch, Fake Happiness detection, and purchase-intent estimation
- Designed a complete pipeline from data preprocessing to evaluation, dissonance reporting, and frontend deployment using saved PyTorch model files

## 6.3 Limitations

- Limited dataset diversity (only three advertisement categories)
- Class imbalance in the biosignal stream (Joy and Surprise dominate over 67% of the data)
- Hash-based linking provides a reproducible pairing rather than recovering the original session-level alignment
- Short average video duration restricts the temporal context available to CNN3D
- System may struggle with underrepresented emotion classes such as Anger, Sadness, and Fear
- Rule-based fusion grouping is informed by affective computing principles but is not learned from data, which may limit generalisation to datasets with different emotion distributions

## 6.4 Future Scope

- Use larger and more diverse multimodal emotion datasets with preserved session-level alignment
- Replace the rule-based fusion layer with cross-attention transformer heads for richer cross-modal interactions when more aligned data becomes available
- Replace dominant-emotion labels with continuous valence-arousal targets for regression-based dissonance scoring
- Incorporate participant demographic information from questionnaires as auxiliary inputs
- Add a user-friendly interface (web or mobile application) for real-time advertisement analysis
- Implement explainability methods such as Grad-CAM for video and saliency maps for biosignals to interpret model decisions
- Extend the affective grouping rule to include additional categories (e.g., separating high-arousal positive from low-arousal positive) for finer-grained dissonance detection

# REFERENCES

| Ref | Paper | Dataset Used | Limitations | Comparison With Our Project | Model Used |
|---|---|---|---|---|---|
| **[1]** | **MVP — Strizhkova et al. (2024) arXiv:2501.03103** | MAHNOB-HCI, AMIGOS, DEAP — face video + EDA + ECG/PPG, sessions 1–2 minutes | Requires long aligned sequences (1–2 min); needs heavy attention-based pretraining; only binary valence/arousal; fails on short clips | Uses a transformer with cross-attention for fusion and relies on session-level alignment; predicts only valence/arousal. Our project uses a simpler CNN3D + 1D-CNN + rule-based fusion, handles 7 emotion classes, and works without session-level alignment via hash-based pairing | VideoMAE (video) + 1D-CNN UniTransformer (physio) + Cross-attention transformer fusion |
| **[2]** | **GBV-Net — Yu, Ru, Lei, Chen (Sensors 2025)** | DEAP and MAHNOB-HCI (EEG, ECG, EOG, GSR + facial video) | Requires multiple aligned physiological channels (EEG, ECG, EOG, GSR) which most wearables don't capture; computationally heavy gating; only valence/arousal | Uses gated attention fusion needing EEG-grade signals. Our project uses simpler Empatica E4 signals (BVP, EDA, TEMP) with a rule-based affective grouping instead of learned gating | ConvNeXt V2 + Transformer (face) + 1D-CNN/TCN/self-attention (physio) + Gated attention fusion |
| **[3]** | **Antonov et al. (Sci. Reports 2024) — Decoding viewer emotions in video ads** | System1 proprietary dataset — 30,000+ video ads with 2.3M annotations across 8 emotion classes by 75 viewers each | No physiological signals — relies only on third-party viewer self-reports; proprietary dataset not public; cannot detect hidden/dissonant emotions | Uses only video + audio, no physiology, so cannot detect emotional dissonance between facial expression and internal physiological state. Our work fuses video AND physiology to detect hidden reactions | TSAM (Temporal Shift Attention Model) — multi-modal CNN with ResNet-50 backbone + audio mel-spectrograms |
| **[4]** | **Ramaswamy & Palaniswamy (WIREs DMKD 2024) — Comprehensive Review** | Reviews 179 multimodal emotion recognition papers (2017–2023): DEAP, MAHNOB-HCI, AMIGOS, IEMOCAP, etc. | This is a survey paper, not an empirical study — no model trained, no benchmark results produced | Survey only; proposes no system. Our project provides an actual implemented pipeline with measured results on the NeuroBioSense dataset | None — pure literature review covering emotion theories, modalities, and fusion strategies |
| **[5]** | **Mocanu, Tapu, Zaharia (Information Fusion 2024) — State-of-the-art MER survey** | Surveys MER datasets across audio, video, EEG, peripheral signals (RAVDESS, IEMOCAP, MELD, etc.) | Survey paper; focuses heavily on audio–video rather than facial-video + biosignal fusion; no original experiments | Their related practical work focuses on audio–video fusion. Our project uses video + physiological fusion (no audio) and adds an interpretable rule-based affective grouping layer | None — survey paper; their related practical work uses cross-modal attention with deep metric learning |

| | | | | | |
|---|---|---|---|---|---|
| [6] | **Carreira & Zisserman (CVPR 2017) — Quo Vadis, Action Recognition? (I3D)** | Kinetics-400 (400 action classes, 400+ clips per class from YouTube) + UCF-101 + HMDB-51 | Designed for action recognition, not emotion recognition; needs huge pretraining datasets; computationally expensive (64-frame clips at 25 fps); not domain-tuned for affective signals | Introduces I3D for action recognition. We use Kinetics-400 pretraining indirectly via R3D-18 weights but apply it to emotion classification on advertisements — a domain shift Carreira/Zisserman never addressed | I3D — 2D Inception-V1 inflated into 3D convolutions |
| [7] | **Tran et al. (CVPR 2018) — A Closer Look at Spatiotemporal Convolutions (R(2+1)D, R3D)** | Kinetics-400 + Sports-1M + UCF-101 + HMDB-51 (action recognition datasets) | Action-recognition benchmark only; no emotion or physiological data; computationally heavy at inference; treats space and time independently | This paper introduces the R3D-18 backbone we directly use as our video encoder, but they use it only for action classification. We fine-tune R3D-18 for 7-class emotion classification on ads and pair it with a biosignal encoder | R3D-18, R(2+1)D, MCx, rMCx — 3D residual networks with various spatio-temporal factorisations |
| [8] | **He et al. (CVPR 2016) — Deep Residual Learning for Image Recognition (ResNet)** | ImageNet (1.28M images, 1000 classes) + CIFAR-10 + COCO | 2D image classification only — completely ignores temporal dimension; no video, no physiology, no emotion; needs full RGB images | Pure 2D image classification — discards temporal information. Our earlier baseline used ResNet-18 on just two frames per video, which is exactly the limitation we replaced by switching to R3D-18 with 16 frames | ResNet-18, ResNet-50, ResNet-101, ResNet-152 — 2D residual CNN with skip connections |
| [9] | **Yin et al. (cited as Ortega et al.) arXiv:1811.07392, 2018 — Facial Expression and Peripheral Physiology Fusion** | Their own small dataset — 12 participants, 4 emotion-induction videos, facial videos + peripheral physiology (EDA, BVP, etc.) | Very small dataset (12 participants); uses hand-crafted recurrence-network features rather than deep learning; relies on personalised resting baselines per subject; not scalable | Uses person-specific recurrence networks and hand-crafted complex network metrics — not deep learning. Our project uses end-to-end CNNs (no per-subject calibration, no hand-crafted features) on a much larger dataset (1,296 ads, 58 participants) | Person-specific recurrence networks + complex network metrics + classical inference models (no deep neural network) |
| [10] | **Bota, Wang, Fred, Plácido da Silva (IEEE Access 2019) — Review on Physiological Emotion Recognition** | Reviews physiological-only datasets — DEAP, MAHNOB-HCI, AMIGOS, ASCERTAIN, WESAD, CASE — and Empatica E4 / wearable studies | Survey paper — no implementation; covers only physiological modality (no video/multimodal); identifies challenges like noise, individual differences, missing data | Survey only. Our project uses one of the wearables they review (Empatica E4) but goes beyond their physiological-only scope by adding video and rule-based fusion | None — survey covers SVM, Random Forest, kNN, CNN, LSTM applied to physiological signals |